👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD MPP.7

Math is Boundless

Lesson 10-5 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

Language Objective: I can describe a design using the words repetition, pattern, pattern unit, and rhythm.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

repetition | **pattern** | **pattern unit** | **rhythm** | **predictability**

Level 3: try the blanks from memory first, then check the bank.

1. The same object or shape repeated multiple times is _____.

2. Objects or shapes repeated in a regular order form a _____.

3. The smallest chunk of a pattern that repeats is the _____ _____ .

4. Varying the placement, size, or order of the same shapes creates _____ .

5. Knowing what comes next because the design follows a rule is _____ .

Write About the Math

How is a design with rhythm different from a pattern?
¿En qué se diferencia un diseño con ritmo de un patrón?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **repetition** — *repetición*
- ☐ **pattern** — *patrón*
- ☐ **pattern unit** — *unidad del patrón*
- ☐ **rhythm** — *ritmo*
- ☐ **predictability** — *previsibilidad*

Model: *The pattern unit is 3 shapes long, and 12 ÷ 3 = 4 with no remainder, so shape 12 ends a complete unit — it is the blue triangle.* — A strong answer divides 12 by 3, gets remainder 0, and explains that means the LAST shape in the unit. A weak answer just guesses the shape without dividing.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The circle painting's colors and sizes ___, which is why it differs from a ___ — you cannot ___ what the next circle will look like.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** repetition
2. **Notes 2:** pattern
3. **Notes 3:** pattern unit
4. **Notes 4:** rhythm
5. **Notes 5:** predictability
6. **Watch for this mistake:** *Calling every repeated design a 'pattern.' One identical shape repeated is repetition; a pattern needs a unit of shapes repeating in a regular order; and rhythm varies the shapes' size and order, so it is neither.*

7. **Try It 1:**
8. **Try It 2:**